

Banking Management System Requirements

Functional Requirements

Admin Functional Requirements

1. The system shall allow the admin to create teller accounts.
2. The system shall allow the admin to view all banking transactions.
3. The system shall allow the admin to view system summaries and statistics.
4. The system shall allow the admin to create branches.
5. The system shall allow the admin to update branch information.
6. The system shall allow the admin to delete branches.
7. The system shall allow the admin to view all tellers.

Teller Functional Requirements

8. The system shall allow the teller to create customer accounts.
9. The system shall allow the teller to create bank accounts for customers.
10. The system shall allow the teller to deposit money into accounts.
11. The system shall allow the teller to withdraw money from accounts.
12. The system shall allow the teller to create loans for customers.
13. The system shall allow the teller to view customer accounts.

Customer Functional Requirements

14. The system shall allow customers to view their accounts.
15. The system shall allow customers to view transaction history.
16. The system shall allow customers to view their loans.
17. The system shall allow customers to transfer money between accounts.
18. The system shall allow customers to check account balances.
19. The system shall allow customers to view total balances across accounts.
20. The system shall allow customers to pay loan installments.
21. The system shall allow customers to view their profiles.

Non-Functional Requirements

Security Requirements

22. The system shall authenticate all users before granting access.
23. The system shall use role-based authorization (Admin, Teller, Customer).
24. Sensitive customer data shall be encrypted.
25. The system shall securely store passwords using hashing techniques.
26. The system shall log all financial transactions for auditing.

Performance Requirements

27. The system shall process transactions within 3 seconds.
28. The system shall support multiple concurrent users.

29. The system shall retrieve account information quickly and efficiently.

Reliability Requirements

30. The system shall ensure data consistency during transactions.

31. The system shall automatically recover from unexpected failures.

32. The system shall maintain backup copies of data regularly.

Usability Requirements

33. The system shall provide a user-friendly interface.

34. The system shall display clear validation and error messages.

35. The system shall support easy navigation between system modules.

Availability Requirements

36. The system shall be available 24/7 except during maintenance periods.

37. The system shall minimize downtime during updates.

Scalability Requirements

38. The system shall support future expansion of branches and users.

39. The system shall handle increasing numbers of transactions efficiently.

Maintainability Requirements

40. The system shall follow clean architecture principles.

41. The system shall allow easy maintenance and feature updates.

42. The system shall keep logs for debugging and monitoring purposes.